Manya Pandey

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Education

Kalinga Institute of Industrial Technology — Bhubaneshwar, India
Bachelor of Technology in Computer Science — *Aug. 2023 – Jul. 2027*

Experience

AI Research Intern — Feb. 2026 – Apr. 2026, Aug. 2026 – Present
Birla AI Labs, Office of Ananya Birla (OAB) — *Mumbai, India (Remote)*

- Identified **context poisoning** in patch-based Time Series Foundation Models (TSFMs), where structurally anomalous patches capture disproportionate attention and silently degrade **zero-shot forecast quality**.
- Designed **GITCO**, a lightweight **inference-time context optimization** framework (Gate–Router–Critic) that selectively suppresses harmful patches **without any parameter updates**, preserving full model generalizability.
- Evaluated on **TimesFM 2.5** across **53 GIFT-Eval datasets** under K-fold cross-validation; achieved **+1.95% average MASE reduction** while capturing **89.9% of the theoretical improvement upper bound**.
- Introduced **context sensitivity profiles** as a novel characterizable property of TSFMs, mapping time series meta-features to expected accuracy gains under inference-time intervention.

Software Development Engineering Intern — May 2026 – Jul. 2026
Bank of New York (BNY) — *Pune, India*

- Engineered an **NLP-to-pandas** search pipeline for **100+ economic stress-testing scenarios**, eliminating manual spreadsheet filtering.
- Designed a 3-tier dynamic LLM context-routing system, **reducing API costs by 6.3x** while maintaining **<2s p95 latency**; hardened it with **13 prompt-injection guardrails**, deterministic generation, and Levenshtein fuzzy-matching.
- Developed a **hybrid RAG retrieval service** over **700+ pages** of regulatory text, integrated via the **Model Context Protocol (MCP)**.
- Architected a **3-signal fusion ranking algorithm** (BM25, Cosine Similarity, Metadata), achieving **81% Recall@5** and **0.63 MRR**, with strict source citation logic from a **ChromaDB** vector store for verifiable financial workflows.

Leadership & Additional Experience

AI/ML Lead, *Google Developer Group (GDG), Bhubaneshwar* — *Aug. 2025 – Present*

Undergraduate Researcher, *RespAI Lab, Bhubaneshwar* — *Mar. 2025 – Present*

Honors: Rank 19 of ∼32,000 teams, **Amazon ML Challenge** 2026 | Selected Delegate, **Google DeepMind APAC Research Symposium** 2026 | **Adobe Student Insider**, 2026-27 Cohort

Publications

GITCO: Gated Inference-Time Context Optimization | *ICML 2026 Workshop* — Jun 2026

- Proposed a novel context optimization framework improving inference-time capabilities of timeseries foundational models; accepted at the **ICML Foundation Models for Structured Data (FMSD)** and **GlobalSouth** workshops.

Confidence Is Not Competence | *Preprint* — Oct 2025

- Identified mechanistic disconnect between confidence and reasoning in LLMs (Llama 3.1, Mistral, Gemma 3) using **linear probes**, **steering vectors**, and **PCA**-based dimensionality analysis.

Certifications

Technical AI Safety – BlueDot Impact
Build and Deploy Machine Learning Solutions on Vertex AI – Google Cloud Skills Boost
MLOps with Vertex AI: Model Evaluation – Google Cloud Skills Boost

Technical Skills

Languages: Java, Python, C, SQL, LaTeX
CS Fundamentals: Object Oriented Programming, Data Structures & Algorithms, Operating System, Git, Cloud Computing, Database Management Systems
ML / Research: PyTorch, TensorFlow, Hugging Face, NumPy, pandas, Scikit-learn, OpenCV, WandB, LLM
Relevant Coursework: Machine Learning, Artificial Intelligence, Deep Learning, Robotics, Intro to LLM, Differential Equations and Linear Algebra, Probability and Statistics